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Bihar Engineering University, Patna

B.Tech 1st Semester Examination, 2024 (Old)

Time: 03 Hours

Code: 100103 ***Subject: Chemistry*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option/answer the following (Any seven only): **[14]**

(a) According to Heisenberg's Uncertainty Principle, which of the following pairs cannot be simultaneously determined with high precision?

(i) Energy and mass
(ii) Position and momentum
(iii) Velocity and acceleration
(iv) Force and time

(b) According to Crystal Field Theory, the splitting of d-orbitals in an octahedral field leads to:

(i) Equal energy for all five d orbitals
(ii) Splitting into two lower and three higher energy orbitals
(iii) Splitting into three lower and two higher energy orbitals
(iv) Two sets of equal energy

(c) Which of the following complexes will show geometrical isomerism?

(i) $[Co(NH_3)_6]^{3+}$
(ii) $[Pt(NH_3)_2Cl_2]$
(iii) $[Cr(en)_3]^{3-}$
(iv) $[Fe(CN)_6]^{3-}$

(d) In IR spectroscopy, a vibration is IR active only if:

(i) There is a change in bond energy
(ii) The dipole moment changes during vibration
(iii) The bond length remains constant
(iv) The mass of atoms is equal

(e) Which of the following is not a van der Waals force?

(i) Dipole-dipole
(ii) Hydrogen bonding
(iii) London dispersion
(iv) Ionic bond

(f) The spontaneity of a chemical process is indicated by:

(i) $\Delta G=0$
(ii) $\Delta G < 0$
(iii) $\Delta G > 0$
(iv) $\Delta S < 0$

(g) Temporary hardness of water is mainly due to:

(i) Calcium bicarbonate
(ii) Magnesium sulfate
(iii) Calcium chloride
(iv) Sodium carbonate

(h) Which of the following elements has the highest electronegativity?

(i) Oxygen
(ii) Chlorine
(iii) Nitrogen
(iv) Fluorine

(i) Enantiomers differ in:

(i) Boiling point
(ii) Refractive index
(iii) Optical rotation
(iv) Melting point

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(j) Diels-Alder reaction involves

(i) Electrophilic addition

(ii) Nucleophilic substitution

(iii) Cycloaddition between a diene and a dienophile

(iv) Radical chain mechanism

Q.2 Solve both questions : **[14]**

(a) State and explain Heisenberg's Uncertainty Principle. Discuss its significance in determining position and momentum of an electron.

(b) Explain the photoelectric effect and de Broglie hypothesis.

Q.3 Solve both questions : **[14]**

(a) Explain the formation of molecular orbitals using the linear combination of atomic orbitals (LCAO) approach. Differentiate between bonding and antibonding orbitals.

(b) Discuss the crystal field splitting of d-orbital in Tetrahedral complexes.

Q.4 Solve both questions : **[14]**

(a) Explain the principles of vibrational and rotational spectroscopy. What are the selection rules for rotational transitions in diatomic molecules?

(b) Derive van der Waal's equation for real gases.

Q.5 Solve both questions : **[14]**

(a) Discuss the deviation of real gases from ideal behavior. Under what conditions is the deviation more significant?

(b) Discuss the basic principle of Nuclear Magnetic Resonance (NMR) spectroscopy. How is it useful for structural elucidation of organic compounds?

Q.6 Solve both questions : **[14]**

(a) Explain the Nernst equation and its applications in calculating EMF of electrochemical cells.

(b) What do you mean by Hardness of water. Discuss the EDTA method for estimation of hardness of water.

Q.7 Solve both questions : **[14]**

(a) Explain the HSAB (Hard and Soft Acids and Bases) principle with suitable examples.

(b) Define ionization energy & electron affinity, How do these vary in the periodic table as the atomic number increases.

Q.8 Solve both questions : **[14]**

(a) Discuss the relationship between hybridization and molecular geometry. Provide examples of common geometries like tetrahedral, linear, and trigonal planar.

(b) Define stereoisomerism. Differentiate between structural isomers, enantiomers, and diastereomers with examples.

Q.9 Solve both questions : **[14]**

(a) Describe the mechanism of epoxide ring opening reactions under acidic and basic conditions.

(b) Outline the synthesis of aspirin from salicylic acid. Include the reagents, reaction mechanism, and significance of the reaction.

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